

Big Picture: From Thermodynamic Laws to Real Fluid Calculations

CBE 20260 / Thermodynamics I

Edward Maginn

Table of contents

1	The Big Picture of the Second Half of the Course	2
1.1	The Problem	2
2	The Big Idea	3
2.1	The Goal	3
3	Step 1: Equations of State (EOS)	3
3.1	Key Insight	4
4	Step 2: The Fundamental Equation	5
4.1	Big Interpretation	5
5	Step 3: Changing Variables (Legendre Transforms)	5
5.1	Result: Four Potentials	6
6	Step 4: Maxwell Relations (The Key Tool)	6
7	Step 5: General Property Equations	6
7.1	Enthalpy	6
7.2	Internal Energy	6
7.3	Big Result	7
8	Step 6: The Roadblock	7
9	Step 7: The Breakthrough — Residual Properties	7
9.1	Why This Works	7
9.2	Master Equations	8
9.3	Big Insight	8
10	Step 8: Equations of State Become Powerful	8
10.1	Examples	8

11 Step 9: Generalized Correlations	9
11.1 Interpretation	9
12 Step 10: Final Toolbox	9
13 Where This Leads	10

1 The Big Picture of the Second Half of the Course

We have been on a journey. We started off by stating some simple postulates and invoked the First and Second Laws. From this, we were able to solve problems involving pumps, compressors, turbines, etc. by writing mass, energy, and entropy balances. We put together these elements and solved more complicated problems:

- Rankine cycles
- Refrigeration cycles

We saw that the First and Second Laws imposed limits on what we could do.

We needed thermodynamic properties to solve these problems, so we used either the ideal gas assumption or tables like the steam tables, NIST, or CoolProp.

These worked well—but only for **specific fluids and tabulated data**.

1.1 The Problem

What if:

- The fluid is new (no tables exist)?
- The pressure is very high?
- The fluid is not water?

You cannot rely on tables. You need a **general method**.

2 The Big Idea

Remember the Fundamental Postulate of Thermodynamics we talked about in Lecture 4:

Isolated systems with no external forces acting on them will approach unique equilibrium states characterized by the mass of each distinct species in the system and two independently variable properties.

So if we know:

$$P, V, T$$

then in principle we can determine:

$$U, H, S, G$$

2.1 The Goal

Build a framework to compute ALL thermodynamic properties from:

- an equation of state (EOS)
 - measurable variables (P, V, T)
-

3 Step 1: Equations of State (EOS)

We began by introducing equations of state:

- Ideal gas:

$$P\underline{V} = RT$$

- More realistic cubic equations of state:

– van der Waals:

$$P = \frac{RT}{\underline{V} - b} - \frac{a}{\underline{V}^2}$$

- Peng–Robinson:

$$P = \frac{RT}{\underline{V} - b} - \frac{a\alpha(T)}{\underline{V}(\underline{V} + b) + b(\underline{V} - b)}$$

- SRK:

$$P = \frac{RT}{\underline{V} - b} - \frac{a\alpha(T)}{\underline{V}(\underline{V} + b)}$$

- There are other EOS:

- Virial EOS:

$$P = \frac{RT}{\underline{V}} \left(1 + \frac{B(T)}{\underline{V}} + \frac{C(T)}{\underline{V}^2} + \dots \right)$$

- The cubic EOSs were motivated by concepts from corresponding states theory, which says that fluids at the same reduced temperature and pressure have similar behavior. The virial EOS was motivated by statistical mechanics and the idea of a cluster expansion.

These EOS all relate:

$$P = f(T, V)$$

3.1 Key Insight

An EOS gives you **P–V–T behavior**, but engineers need:

- enthalpy
- entropy
- internal energy

□ So we need a way to connect **PVT → all properties**

4 Step 2: The Fundamental Equation

From the First + Second Laws for a closed system:

$$dU = TdS - PdV$$

This is the **starting point of everything**.

It tells us:

- $U = U(S, V)$
 - $T = \left(\frac{\partial U}{\partial S}\right)_V$
 - $P = -\left(\frac{\partial U}{\partial V}\right)_S$
-

4.1 Big Interpretation

Temperature and pressure are **slopes of a thermodynamic surface**

5 Step 3: Changing Variables (Legendre Transforms)

We don't want to work in (S, V) .

We want:

- T and P (measurable!)

So we transform:

- $U \rightarrow H = U + PV$
 - $U \rightarrow A = U - TS$
 - $U \rightarrow G = U + PV - TS$
-

5.1 Result: Four Potentials

Each has its own “natural variables”:

Potential	Natural Variables
U	(S, V)
H	(S, P)
A	(T, V)
G	(T, P)

6 Step 4: Maxwell Relations (The Key Tool)

From exact differentials:

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$$

These allow us to:

Replace **unmeasurable derivatives (entropy)** with **measurable ones (PVT)**

7 Step 5: General Property Equations

Using calculus + Maxwell relations, we derived:

7.1 Enthalpy

$$dH = C_p dT + \left[V - T \left(\frac{\partial V}{\partial T} \right)_P \right] dP$$

7.2 Internal Energy

$$dU = C_v dT + \left[T \left(\frac{\partial P}{\partial T} \right)_V - P \right] dV$$

7.3 Big Result

If you know an EOS (i.e., $V = f(T, P)$), you can compute H , U , and S .

8 Step 6: The Roadblock

In real problems:

$$\Delta H = \int C_p(T, P) dT + \text{complicated term}$$

But:

- We only have **ideal gas** $C_p(T)$ data
 - EOS integrals are messy
- This is where students usually get lost.
-

9 Step 7: The Breakthrough — Residual Properties

Define:

$$M^R = M - M^{IG}$$

So:

$$M = M^{IG} + M^R$$

9.1 Why This Works

We split the problem:

1. Ideal gas part → easy (tables or C_p correlations)
 2. Residual part → from EOS
-

9.2 Master Equations

$$\Delta H = \Delta H^{IG} + (H_2^R - H_1^R)$$

$$\Delta S = \Delta S^{IG} + (S_2^R - S_1^R)$$

9.3 Big Insight

EOS gives you the **correction to ideal gas behavior**

:contentReferenceoacite:2

10 Step 8: Equations of State Become Powerful

Now EOS is not just:

$$P = f(T, V)$$

It becomes a tool to compute:

- Z
 - H^R
 - S^R
 - U^R
-

10.1 Examples

- van der Waals
 - Peng–Robinson
 - SRK
-

11 Step 9: Generalized Correlations

Instead of solving EOS every time:

- Use reduced properties:

$$T_r = \frac{T}{T_c}, \quad P_r = \frac{P}{P_c}$$

- Use acentric factor ω

Then:

$$Z = Z^{(0)} + \omega Z^{(1)}$$

:contentReferenceoaicite:3

11.1 Interpretation

Many fluids behave similarly when scaled properly

12 Step 10: Final Toolbox

At this point, you can solve:

- Compressors (real gases)
- Turbines
- Heat exchangers
- Phase equilibria

WITHOUT tables.

Everything we have done was about replacing steam tables with equations you can apply to *any* fluid.

What You Should Understand (Not Memorize) - Conceptual

- Why 2 properties define a state
- Why ideal gas is only an approximation
- How to compute ΔH , ΔS for real fluids
- How EOS connects to real calculations
- Why residual properties exist

Final Takeaway

-Everything in this section is about one idea:

- Using measurable variables (P, V, T) to compute everything else

-This is what allows chemical engineers to:

- design processes
- simulate systems
- predict behavior of new fluids

13 Where This Leads

You are now ready for:

- Phase equilibria
- Fugacity
- Real mixture thermodynamics

These are the tools used in:

- distillation
- separations
- reactors